DemoMesh: Residual Diffusion for Missing Sensor Readings

I. METHOD

This is an original synthetic document for software demonstrations. DemoMesh and ToyGrid are fictional. No empirical research results or published scientific claims are presented.

DemoMesh fills missing sensor readings in two stages. First, linear interpolation creates a preliminary estimate from observed timestamps. Second, a residual diffusion model learns a correction using neighboring sensors. The final reading is the preliminary estimate plus the predicted residual.

Only observed readings and their masks are supplied to the encoder. Missing readings are excluded from the reconstruction loss. A graph connects spatial neighbors to provide local context. The model does not use future readings during evaluation.

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II. DATA AND LIMITATIONS

ToyGrid is a fictional demonstration dataset with four sensors and twelve timestamps. It exists only to explain the data flow; it is not a benchmark downloaded from a real deployment.

For the example evaluation, the input mask hides selected readings while the held-out values remain available to the evaluator. Mean absolute error compares the predictions with those held-out values. No numerical accuracy result is claimed.

The demonstration assumes synchronized sensors. Clock drift, sensor failures and adaptation to a new city are outside its scope. This document contains no medical evidence and cannot support advice about insomnia, medication or treatments.